

0 9

Cystic fibrosis (CF) is a recessive inherited disorder caused by a defective *CFTR* gene.

Table 6 shows the number of people with CF and the estimated population of the UK in 2020 and 2022.

Table 6

Year	Number of people with CF	Estimated population of the UK $\times 10^6$
2020	10 837	67.06
2022	11 148	67.51

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Calculate how many more people had CF than was expected in 2022.

You can assume the relative frequency of CF in a population remains constant.

Show your working.

[2 marks]

Additional number of people _____

Question 9 continues on the next page

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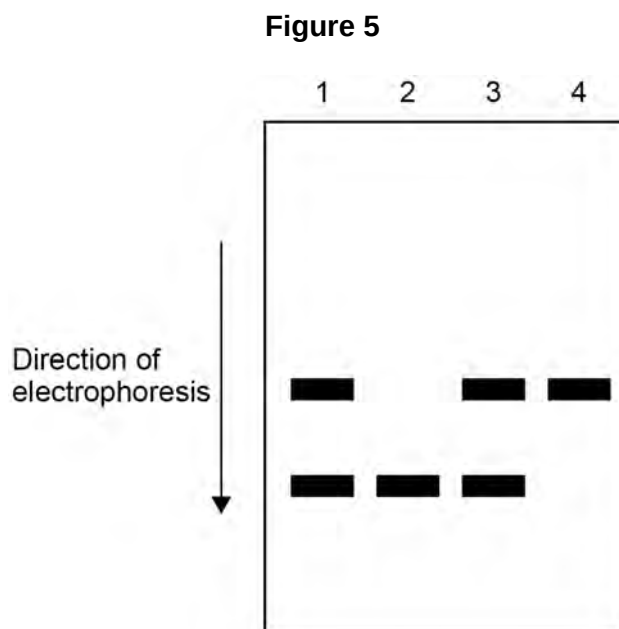
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ANSWER IN THE SPACES PROVIDED

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CF can be caused by a three-base deletion mutation in the *CFTR* gene. This deletion creates a recessive allele called F508del.

DNA probes and gel electrophoresis were used to screen two parents and their two children for the F508del allele. One of the children was diagnosed with CF.

Figure 5 shows the results.



0 9 . 2 What is a DNA probe?

[2 marks]

0 9 . 3

The identity of each family member can be determined from the relative position of the bands shown in **Figure 5**.

Use information in **Figure 5** to:

- complete **Table 7**
- explain how you identified the child with CF.

You can use **F** to represent the dominant *CFTR* allele and **f** to represent the recessive F508del *CFTR* allele.

[3 marks]

Table 7

Lanes	Parent or child	Genotype
1		
2		
3		
4		

Explanation _____

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